GENETICS CO., LTD.

Company Profile

Dairy Embryo Production, Embryo Transfer and
Korean Holstein Sexed Embryo Export

Prepared for the Ministry of Economy and Finance of the Republic of Uzbekistan

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Genetics Co., Ltd. · Korea Animal Hospital
Anseong, Gyeonggi-do, Republic of Korea

1. Genetics at a Glance

Genetics Co., Ltd. is a Korean livestock biotechnology company specialised in **dairy embryo production, embryo transfer and reproduction management**. Together with its affiliated veterinary hospital, Korea Animal Hospital, the company produces and transfers Holstein embryos for Korean dairy farms and exports **Korean Holstein sexed (female) dairy embryos** to Central Asia.

Key figures

- Established in **2005**, based in Anseong, Gyeonggi-do – the centre of Korea's dairy industry
- About **3,000 embryo transfers** and about **4,000 embryos sold** per year
- Conception rate of about **60%** for our sexed, frozen IVF embryos transferred abroad over the last five years (company field data)
- Embryo export and on-site transfer support in **Mongolia, Uzbekistan and Kyrgyzstan**
- About three years of Korean Holstein sexed embryo supply to **Uzbekistan** in cooperation with Korea's **Rural Development Administration (RDA)**

2. Company Overview

Item	Details
Company	Genetics Co., Ltd. (Republic of Korea)
Affiliate	Korea Animal Hospital – veterinary hospital specialised in embryo transfer and reproduction
Established	2005
Location	29-67 Yanghyeop-gil, Bogae-myeon, Anseong-si, Gyeonggi-do, Republic of Korea
CEO	Dr. Hyunjae Ha, DVM
Veterinary team	Dr. Hyunjae Ha, Dr. Kwangtaek Lim, Dr. Donghwan Shin, Dr. Mincheol Kim, supported by laboratory researchers
Core business	In-vitro embryo production (IVF), sexed embryos, embryo transfer, genomic selection, recipient-cow programmes, reproduction consulting, embryo export
Related businesses	Song Young Shin Farm (A2A2 Jersey dairy and dairy processing); precision-livestock monitoring with rumen sensors (CowTalk AI); livestock environment technology (D2O)

Founder and CEO – Dr. Hyunjae Ha, DVM

- Doctor of Veterinary Medicine, Konkuk University (1999); Adjunct Professor, Konkuk University.
- More than 25 years in large-animal veterinary practice and 18+ years in embryo transfer and in-vitro fertilisation.
- Breeder of A2A2 Jersey cattle and operator of an organic dairy farm for over 15 years.
- First to introduce smaXtec rumen-sensor monitoring to Korean dairy farms.
- Hands-on experience of embryo export and local transfer programmes in Central Asia (Mongolia, Uzbekistan, Kyrgyzstan).

3. What We Do

Business area	Description
Embryo production	In-vitro production of Holstein embryos fertilised with sexed semen; grading, freezing and inventory management in our own laboratory
Embryo transfer	Transfer of fresh and frozen embryos on dairy farms by our veterinary team

Business area	Description
Recipient programmes	Recipient selection, oestrus synchronisation and pregnancy diagnosis protocols
Genetic improvement	Genomic selection and long-term breeding plans for dairy herds
Embryo export	Export of Korean Holstein sexed dairy embryos with health certificates and embryo identification records
Training and consulting	Training of local veterinarians and technicians in embryo transfer, thawing, pregnancy diagnosis and heifer rearing
Digital follow-up	Herd monitoring with rumen sensors and data platform (CowTalk AI) for reproduction and health management

4. Production Technology and Quality Control

Step	What we do	Why it matters
Oocyte source	Oocytes collected from ovaries of Korean Holstein cows	Large-scale, cost-efficient embryo production from the Korean Holstein population
Fertilisation	In-vitro fertilisation with **sexed semen** from selected sires	About 90% female calves expected
Individual culture	Each embryo is cultured **individually**, not in pooled batches	Every embryo keeps its dam and sire identity – full traceability for breeding records
Grading	Embryos graded by stage and quality following IETS guidance	Only transferable-grade embryos are frozen and exported
Freezing	**Slow freezing in ethylene glycol for direct transfer**	Thawed and transferred on the farm like a semen straw – simple for field technicians
Documentation	Embryo ID list, donor/sire records, health certificate per the importing country's requirements	Supports import approval and national breeding records

Field results depend strongly on recipient management. Our ~60% conception rate abroad was achieved with well-managed recipients; our recommended recipient is a **maiden heifer of breeding age** in good body condition.

5. Why Korean Holstein Sexed Embryos

- **Proven genetics:** the Korean Holstein population has been improved for decades with top sires from North America and Europe and national milk recording.
- **Climate fit:** Korean herds are raised in intensive housing under hot summers and cold winters – conditions comparable in many respects to Uzbekistan's continental climate.
- **Female calves:** sexed semen means mainly female calves – the replacement heifers dairy farms need.
- **Lower biosecurity risk:** embryos handled to WOAH/IETS standards carry a much lower risk of disease transmission than live animals.
- **Efficient logistics:** thousands of embryos travel in one liquid-nitrogen container on a regular flight, without the cost, stress and losses of live-animal transport.
- **Local benefit:** calves are born in the importing country, local heifers are used as recipients, and local veterinarians gain new skills.

6. International Experience

Country	Activities
Uzbekistan	Korean Holstein sexed embryo supply for about three years in cooperation with RDA; on-site transfer support; smart-farming rumen-sensor pilot on about 50 cows
Mongolia	Dairy embryo export and on-site transfer support
Kyrgyzstan	Dairy embryo export and on-site transfer support

Contact

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